

# Classical Mechanics

Lecture 8: Gauge Invariance, Cyclotron Motion, and Poisson Brackets

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## Chapter 1

# Gauge Invariance, Cyclotron Motion, and Poisson Brackets

A charged particle in an electromagnetic field is a small system with an unusually rich canonical structure. Newton's equation is simple, but the momentum that enters Hamiltonian mechanics is not merely  $m\mathbf{v}$ . The action requires a vector potential that is not unique. A uniform magnetic field turns translation invariance into a statement about the center of a cyclotron orbit, and an added electric field produces a steady drift perpendicular to the applied force.

We shall develop those facts in the order in which they arise. The final part of the chapter changes language once more, from Hamilton's equations to Poisson brackets. That algebraic formulation is more abstract than Newton's law, but it is the form that most clearly anticipates quantum mechanics.

### 1.1 Four Forms of the Same Mechanics

For the present discussion the electromagnetic field is prescribed. It may vary from place to place, but it does not depend explicitly on time, and we do not yet solve equations for the field itself. The only dynamical object is a particle of mass  $m$  and charge  $q$ .

It is useful to place four familiar descriptions of mechanics side by side:

1. Newton's equations of motion;
2. the principle of stationary action;
3. the Euler–Lagrange equations;
4. Hamilton's equations in phase space.

They are equivalent when each is available, but they expose different features of a theory. An equation such as  $F = ma$  tells us how a trajectory moves. It does not, by its form alone, tell us why energy or momentum is conserved. The action is deeper, but one normally uses it by taking a variation and obtaining differential equations. The Lagrangian is powerful and direct. The Hamiltonian is often the least intuitive, yet it displays the flow of whole ensembles through phase space, makes information conservation visible, and leads naturally toward quantum mechanics.

The charged particle will make both the power and the strangeness of these descriptions concrete. We restrict the motion to the  $x$ - $y$  plane. The magnetic field is perpendicular to that plane, while an electric field may lie within it.

## 1.2 The Lorentz Force and the Potentials

The nonrelativistic equation of motion is

$$m\ddot{\mathbf{x}} = q\mathbf{E} + q\dot{\mathbf{x}} \times \mathbf{B}. \quad (1.1)$$

We use an electromagnetic convention in which no explicit factor of  $c$  appears. Other conventions insert  $1/c$  in the magnetic term; that change does not alter the argument.

The magnetic force depends on velocity, but it is perpendicular to velocity:

$$\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0. \quad (1.2)$$

Consequently

$$\frac{d}{dt} \left( \frac{1}{2}mv^2 \right) = \mathbf{v} \cdot q(\mathbf{v} \times \mathbf{B}) = 0. \quad (1.3)$$

A magnetic field bends a trajectory without changing its speed. This is the precise content of the statement that magnetic fields do no work. It also explains why planar motion remains planar: with  $\mathbf{v}$  in the plane and  $\mathbf{B}$  perpendicular to it,  $\mathbf{v} \times \mathbf{B}$  lies in the plane.

To pass from the force law to an action, we need potentials. For a region in which the magnetic field has no monopole source,

$$\nabla \cdot \mathbf{B} = 0, \quad (1.4)$$

we may write, at least locally,

$$\boxed{\mathbf{B} = \nabla \times \mathbf{A}.} \quad (1.5)$$

The local qualification matters in topologically nontrivial settings and in theories with magnetic monopoles. For the ordinary fields considered here, one vector potential is sufficient.

**Question from the audience.** Why does writing  $\mathbf{B}$  as a curl imply that magnetic charge is absent?<sup>a</sup>

**Response.** The divergence of every curl vanishes:  $\nabla \cdot (\nabla \times \mathbf{A}) = 0$ . Electric field lines may begin or end on electric charge because  $\nabla \cdot \mathbf{E}$  is proportional to charge density. In the classical electromagnetic theory being used here, there is no corresponding magnetic charge density. Magnetic monopoles are a possible extension of the theory, but none is part of this problem.

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<sup>a</sup>Lecture timestamp: 00:11:34–00:13:18.

The electrostatic field is represented by a scalar potential,

$$\mathbf{E} = -\nabla V, \quad U = qV. \quad (1.6)$$

Thus  $V$  is energy per unit charge, while  $U$  is the potential energy of this particular particle. Doubling  $q$  doubles both the electric force and the potential energy in a fixed external field.

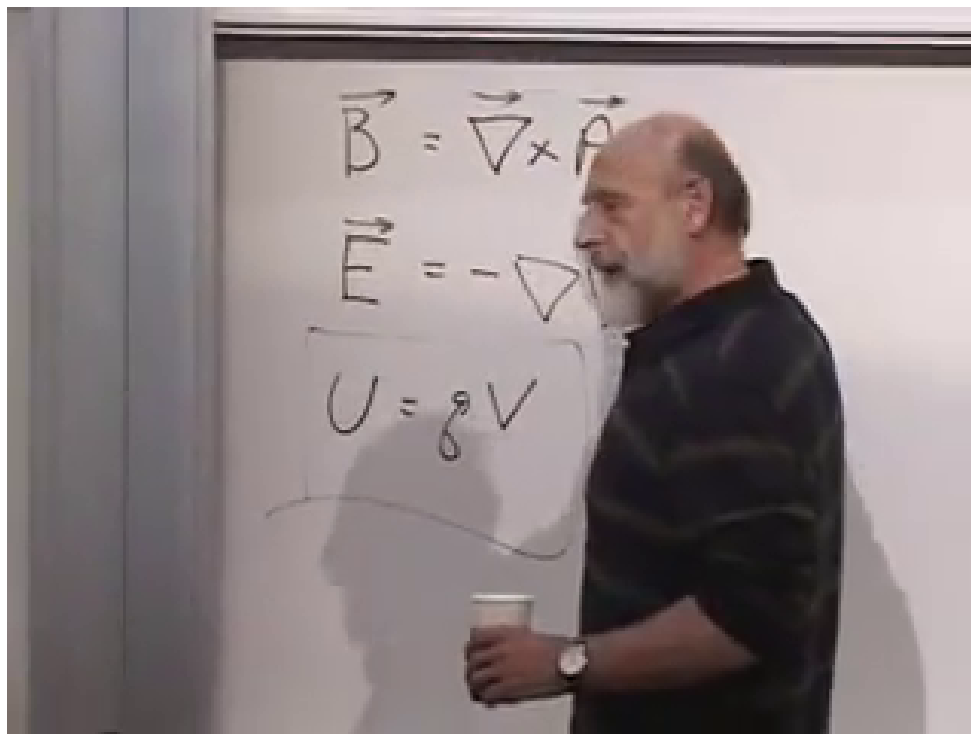


Figure 1.1: The vector potential, the electric potential, and the force law at 00:17:19. The action requires  $\mathbf{A}$ , although the Newtonian equation can be written entirely in terms of  $\mathbf{B}$ .

**Question from the audience.** What does the symbol  $V$  measure? Is it already the particle's potential energy?<sup>a</sup>

**Response.**  $V$  is the electric potential, measured in volts, or energy per unit charge. The particle's potential energy is  $U = qV$ . This distinction keeps the prescribed field separate from the charge used to probe it.

<sup>a</sup>Lecture timestamp: 00:15:43–00:16:31.

### 1.3 The Electromagnetic Action and Gauge Redundancy

For planar motion, the action is

$$S = \int \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) dt + q \int_{\gamma} A_i dx_i - q \int V dt. \quad (1.7)$$

The middle term is a line integral along the path  $\gamma$ . Divide and multiply its infinitesimal displacement by  $dt$ :

$$A_i dx_i = A_i \dot{x}_i dt = \mathbf{A} \cdot \mathbf{v} dt. \quad (1.8)$$

The action therefore has the ordinary form  $S = \int L dt$ , with

$$L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) + q(A_x \dot{x} + A_y \dot{y}) - qV(x, y). \quad (1.9)$$

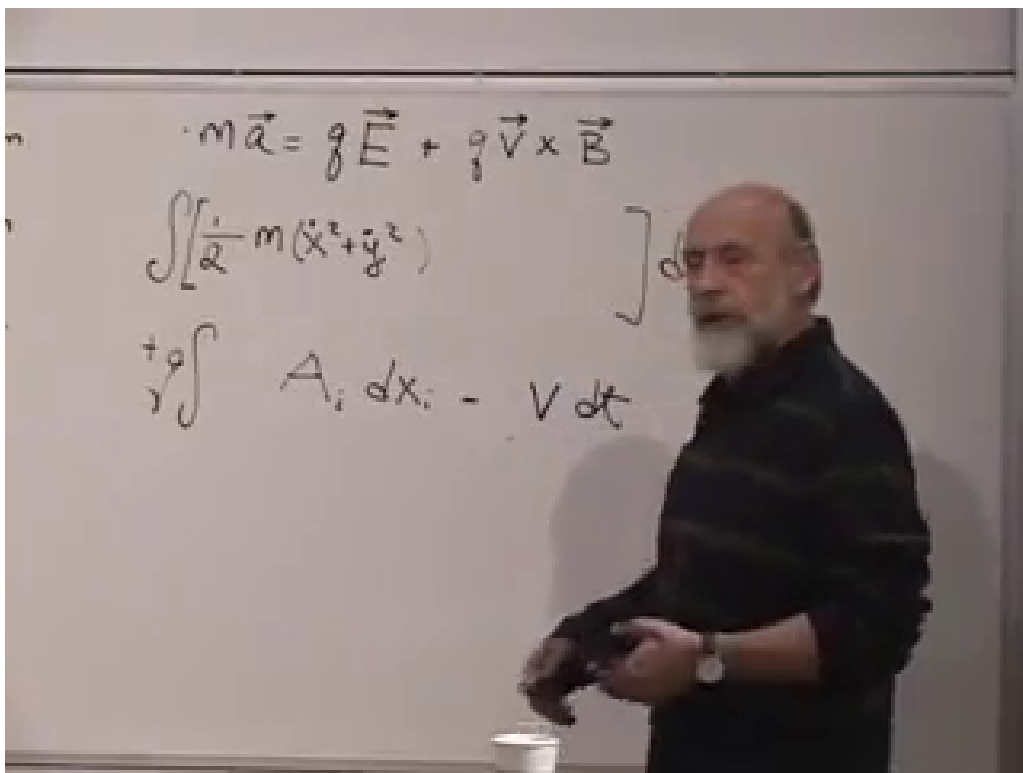


Figure 1.2: The electromagnetic action at 00:22:37. The magnetic coupling is a line integral  $q \int \mathbf{A} \cdot d\mathbf{x}$ , which becomes  $q\mathbf{A} \cdot \mathbf{v}$  in the Lagrangian.

The combination  $A_i dx_i - V dt$  hints at a four-dimensional electromagnetic potential. We do not need special relativity here, but the formula is already arranging the scalar and vector potentials into the form they will take in space-time.

**Question from the audience.** Is the repeated index in  $A_i dx_i$  a sum, and is the same true of the scalar-potential term?<sup>a</sup>

**Response.** Yes for the vector term:  $A_i dx_i = A_x dx + A_y dy$  in the plane. The quantity  $V$  is a scalar, so there is no component sum in  $V dt$ . To avoid confusing electric potential  $V$  with velocity, we shall write the latter as  $\mathbf{v}$ .

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<sup>a</sup>Lecture timestamp: 00:23:36–00:24:49.

The action has introduced more description than the force law seemed to need. The reason is that  $\mathbf{A}$  is not unique. For any smooth scalar function  $\lambda(\mathbf{x})$ ,

$$\mathbf{A} \longrightarrow \mathbf{A}' = \mathbf{A} + \nabla \lambda \quad (1.10)$$

leaves the magnetic field unchanged, since mixed partial derivatives commute:

$$\nabla \times \mathbf{A}' = \nabla \times \mathbf{A} + \nabla \times \nabla \lambda = \mathbf{B}. \quad (1.11)$$

In the time-independent setting, the Lagrangian changes by a total derivative,

$$L' = L + q \nabla \lambda \cdot \dot{\mathbf{x}} = L + q \frac{d\lambda}{dt}. \quad (1.12)$$

Hence

$$S' = S + q\lambda(\mathbf{x}_f) - q\lambda(\mathbf{x}_i). \quad (1.13)$$

The endpoint contribution does not affect a variation with fixed endpoints. The Lagrangian changes, but the trajectory does not. This is gauge invariance: different mathematical descriptions represent the same physical field.

The redundancy is not a defect that can simply be discarded. We could stay with the Lorentz-force equation and never mention  $\mathbf{A}$ , but the local action, canonical momentum, and Hamiltonian all require it. Modern physics often reaches its most useful formulations by introducing such redundant variables and then identifying the transformations that leave observables unchanged.

## 1.4 Recovering the Lorentz Force

We should not accept the Lagrangian merely because it has the right-looking symbols. Let us derive the  $x$ -component of the equation of motion. First,

$$p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + qA_x. \quad (1.14)$$

This is our first warning that canonical momentum and mechanical momentum are different:

$$\mathbf{p}_{\text{can}} = m\mathbf{v} + q\mathbf{A}, \quad \mathbf{p}_{\text{mech}} = m\mathbf{v}. \quad (1.15)$$

The canonical momentum changes when  $\mathbf{A}$  changes gauge. It is a phase space coordinate adapted to the chosen Lagrangian, not by itself a gauge-invariant observable.

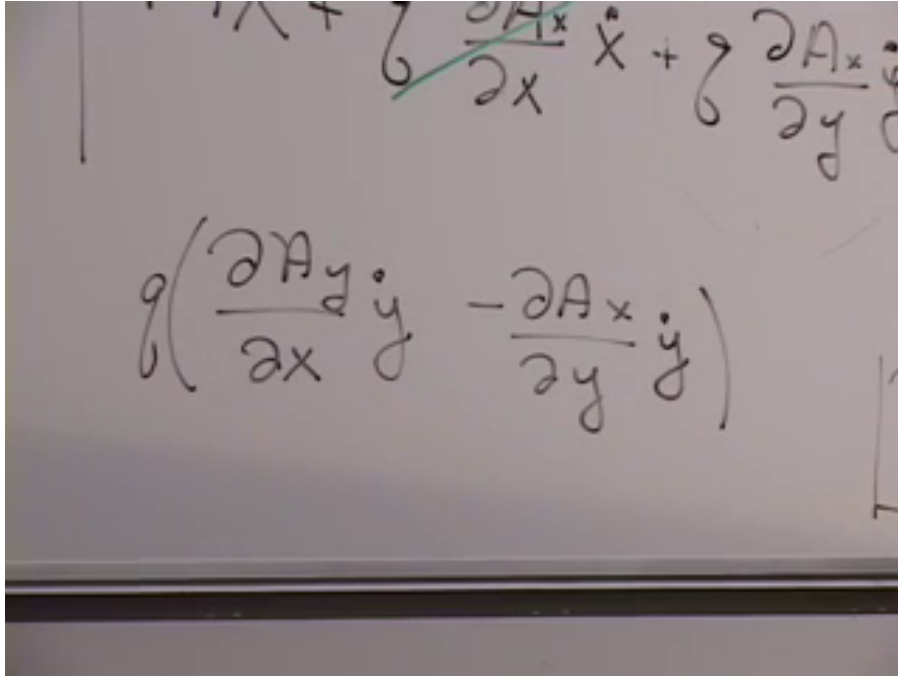


Figure 1.3: The surviving curl combination at 00:38:20. The separate derivatives of  $\mathbf{A}$  combine into the gauge-invariant field  $B_z$ .

The Euler–Lagrange equation gives

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x}. \quad (1.16)$$

Although  $A_x$  has no explicit time dependence, its value at the moving particle changes. The chain rule gives

$$\frac{dA_x}{dt} = \frac{\partial A_x}{\partial x} \dot{x} + \frac{\partial A_x}{\partial y} \dot{y}. \quad (1.17)$$

Therefore

$$m\ddot{x} + q \frac{\partial A_x}{\partial x} \dot{x} + q \frac{\partial A_x}{\partial y} \dot{y} = q \frac{\partial A_x}{\partial x} \dot{x} + q \frac{\partial A_y}{\partial x} \dot{y} - q \frac{\partial V}{\partial x}. \quad (1.18)$$

The common terms cancel, leaving

$$\begin{aligned} m\ddot{x} &= q \left( \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \dot{y} - q \frac{\partial V}{\partial x} \\ &= q B_z \dot{y} + q E_x. \end{aligned} \quad (1.19)$$

The vector potential has disappeared from the final equation except through its curl. Thus the Lagrangian uses a redundant variable, while its Euler–Lagrange equation returns the gauge-invariant force law.

## 1.5 A Uniform Magnetic Field

We now use the canonical formalism rather than merely checking it. Fix ordinary right-handed axes with  $+\hat{z}$  pointing out of the board, and write the magnetic field into the board as

$$\mathbf{B} = -B\hat{z}, \quad B > 0. \quad (1.20)$$

Then  $B_z = -B$ . Two convenient vector potentials for this same field are

$$\text{gauge I:} \quad A_x = By, \quad A_y = 0, \quad (1.21)$$

$$\text{gauge II:} \quad A_x = 0, \quad A_y = -Bx. \quad (1.22)$$

Indeed, each gives  $\partial_x A_y - \partial_y A_x = -B$ . A symmetric choice,

$$A_x = \frac{B}{2}y, \quad A_y = -\frac{B}{2}x, \quad (1.23)$$

does the same. No experiment can distinguish these gauges.

Set the electric field to zero and begin with gauge I. The Lagrangian is

$$L_1 = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) + qBy\dot{x}, \quad (1.24)$$

and the canonical momenta are

$$p_x = m\dot{x} + qBy, \quad p_y = m\dot{y}. \quad (1.25)$$

**Question from the audience.** Which of these two canonical momenta is conserved? <sup>a</sup>

**Response.** The Lagrangian depends on  $y$  but not on  $x$ . A translation  $x \mapsto x + \epsilon$  changes no term appearing in  $L_1$ , so  $x$  is cyclic and  $p_x$  is conserved. A translation in  $y$  changes the term  $qBy\dot{x}$ , so  $p_y = m\dot{y}$  is not the conserved canonical quantity in this gauge.

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<sup>a</sup>Lecture timestamp: 00:46:18–00:48:38.

Choose the conserved value  $p_x = 0$  for the moment. Then

$$\boxed{\dot{x} = -\frac{qB}{m}y.} \quad (1.26)$$

Now use gauge II. Its Lagrangian is

$$L_2 = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - qBx\dot{y}, \quad (1.27)$$

and  $y$  is cyclic. Thus

$$p_y = m\dot{y} - qBx \quad (1.28)$$

is conserved. Choosing  $p_y = 0$  gives

$$\boxed{\dot{y} = \frac{qB}{m}x.} \quad (1.29)$$

It may seem peculiar to use one gauge to expose the  $x$ -relation and another to expose the  $y$ -relation. What is being combined is not two incompatible descriptions at one instant, but two gauge-independent statements about the same physical orbit. Each gauge makes a different translation symmetry manifest.



$$m\vec{a} = q\vec{E} + q\vec{v} \times \vec{B}$$

$$\rightarrow \left[ \frac{1}{2} m(\dot{x}^2 + \dot{y}^2) + qb y \dot{x} \right]$$

$$p_x = m\dot{x} + qby$$

Figure 1.4: Gauge I at 00:46:20. The Lagrangian contains  $qBy\dot{x}$ , so the canonical momentum is  $p_x = m\dot{x} + qBy$ .

$$m\dot{x} + qby = 0$$

$$\boxed{\dot{x} = -\frac{qb}{m} y}$$

Figure 1.5: The first velocity relation at 00:50:10, obtained from the conserved canonical momentum in gauge I.

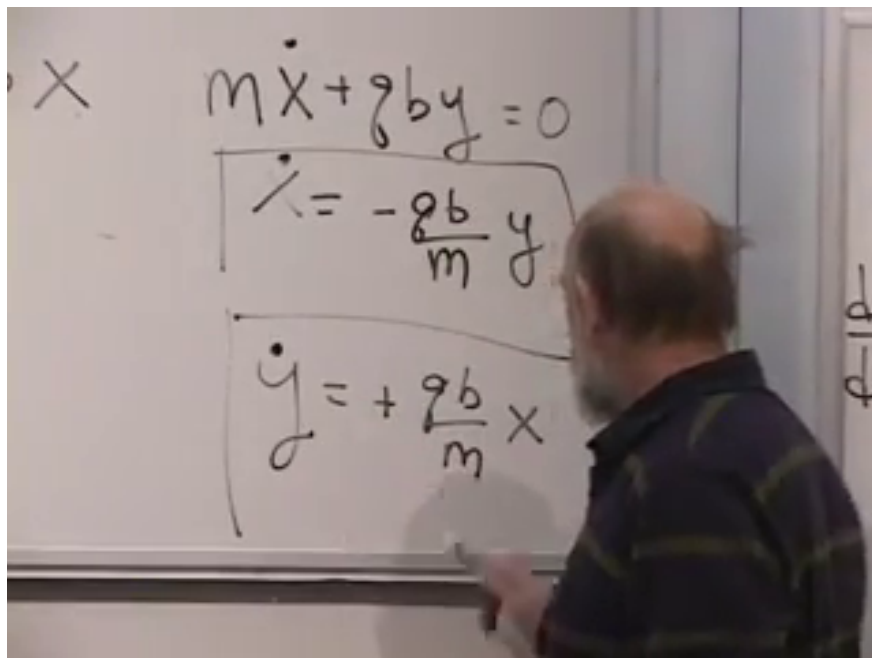


Figure 1.6: The two velocity relations together at 00:52:00. Their form already identifies uniform circular motion.

### 1.5.1 Cyclotron motion and the charge-to-mass ratio

The pair of first-order equations is solved by

$$x = r \cos(\omega_c t), \quad y = r \sin(\omega_c t), \quad (1.30)$$

because

$$\dot{x} = -\omega_c y, \quad \dot{y} = \omega_c x. \quad (1.31)$$

Comparison with Eqs. (1.26) and (1.29) gives the signed cyclotron frequency

$$\boxed{\omega_c = \frac{qB}{m}}, \quad |\omega_c| = \frac{|q|B}{m}. \quad (1.32)$$

The sign fixes the direction of rotation; the magnitude fixes the number of radians swept out per unit time. A larger charge bends the trajectory more rapidly, while a larger mass bends it more slowly.

Historically, magnetic deflection experiments naturally measured  $q/m$ . The trajectory alone did not separately reveal the electron's charge and mass; another kind of measurement was needed to disentangle them.

**Question from the audience.** Does the sign of the cyclotron frequency matter? <sup>a</sup>

**Response.** Its magnitude is  $|q|B/m$ . The sign records orientation. Reversing the charge or the magnetic field reverses the direction in which the orbit is traversed, without changing its radius or speed.

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<sup>a</sup>Lecture timestamp: 00:54:19–00:55:24.

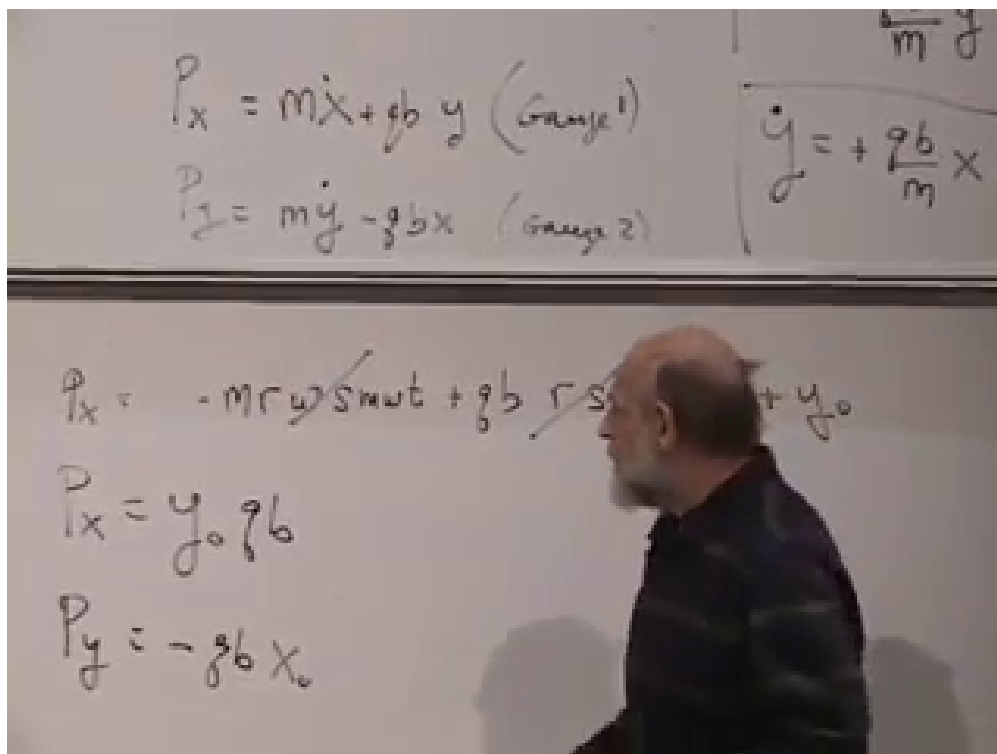


Figure 1.7: The shifted circular orbit and its conserved canonical momenta at 01:01:20. The board gives  $p_x = qBy_0$  and  $p_y = -qBx_0$ .

### 1.5.2 What the conserved momenta measure

A uniform field does not privilege the origin, so the circle may be centered at any fixed point  $(x_0, y_0)$ :

$$x = x_0 + r \cos(\omega_c t), \quad y = y_0 + r \sin(\omega_c t). \quad (1.33)$$

Substitute this orbit into the two conserved canonical momenta. In gauge I,

$$\begin{aligned} p_x &= m\dot{x} + qBy \\ &= -m\omega_c r \sin(\omega_c t) + qB[y_0 + r \sin(\omega_c t)] \\ &= qBy_0. \end{aligned} \quad (1.34)$$

In gauge II,

$$\begin{aligned} p_y &= m\dot{y} - qBx \\ &= m\omega_c r \cos(\omega_c t) - qB[x_0 + r \cos(\omega_c t)] \\ &= -qBx_0. \end{aligned} \quad (1.35)$$

The oscillating terms cancel because  $m\omega_c = qB$ . The strange conserved momenta have a simple physical meaning: they encode the fixed guiding center of the orbit.

**Question from the audience.** Are the dimensions of  $qBy_0$  really those of momentum? <sup>a</sup>

**Response.** Yes, provided one uses one electromagnetic unit convention consistently. In the convention of the present Lagrangian,  $q\mathbf{A}$  has the dimensions of momentum. Since a uniform-field potential scales as  $A \sim B \times \text{length}$ , the product  $qBy_0$  has the same dimensions as  $m\dot{x}$ .

<sup>a</sup>Lecture timestamp: 01:02:19–01:02:45.

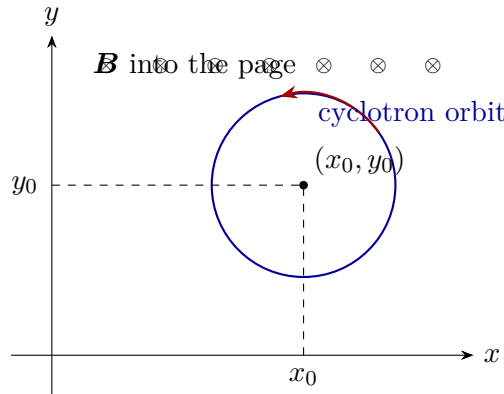


Figure 1.8: Reconstruction of the geometry used at 00:57:52–01:01:35. The canonical constants identify the orbit center rather than the instantaneous mechanical momentum.

## 1.6 Electric Drift and the Hall Response

Restore a constant electric field in the  $+x$ -direction. Choose

$$V(x) = -Ex, \quad \mathbf{E} = E\hat{\mathbf{x}}. \quad (1.36)$$

With the field convention in Eq. (1.20), the component equations are

$$m\ddot{x} + qB\dot{y} = qE, \quad (1.37)$$

$$m\ddot{y} - qB\dot{x} = 0. \quad (1.38)$$

The electric potential depends on  $x$ , so  $x$ -translation invariance is lost and  $p_x$  is no longer conserved:

$$\dot{p}_x = \frac{\partial L}{\partial x} = qE. \quad (1.39)$$

Rather than solve the most general orbit, look first for a trajectory with no acceleration. Setting  $\ddot{x} = \ddot{y} = 0$  in Eqs. (1.37)–(1.38) gives

$$\dot{x} = 0, \quad \boxed{\dot{y} = \frac{E}{B}}. \quad (1.40)$$

The coordinate-free form is

$$\boxed{\mathbf{v}_D = \frac{\mathbf{E} \times \mathbf{B}}{B^2}}. \quad (1.41)$$

For  $\mathbf{E} = E\hat{\mathbf{x}}$  and  $\mathbf{B} = -B\hat{\mathbf{z}}$ , this vector points along  $+\hat{\mathbf{y}}$ . The charge cancels: particles of different charge magnitude share the same  $\mathbf{E} \times \mathbf{B}$  drift. The full crossed-field orbit is a cyclotron motion whose guiding center moves with  $\mathbf{v}_D$ .

With only a magnetic field, the no-acceleration solution is the zero-radius orbit: the particle simply sits still. Turning on  $\mathbf{E}$  changes that solution into steady transverse motion. A distribution of charges therefore develops a sideways flow. This is the elementary mechanical structure behind the Hall response.

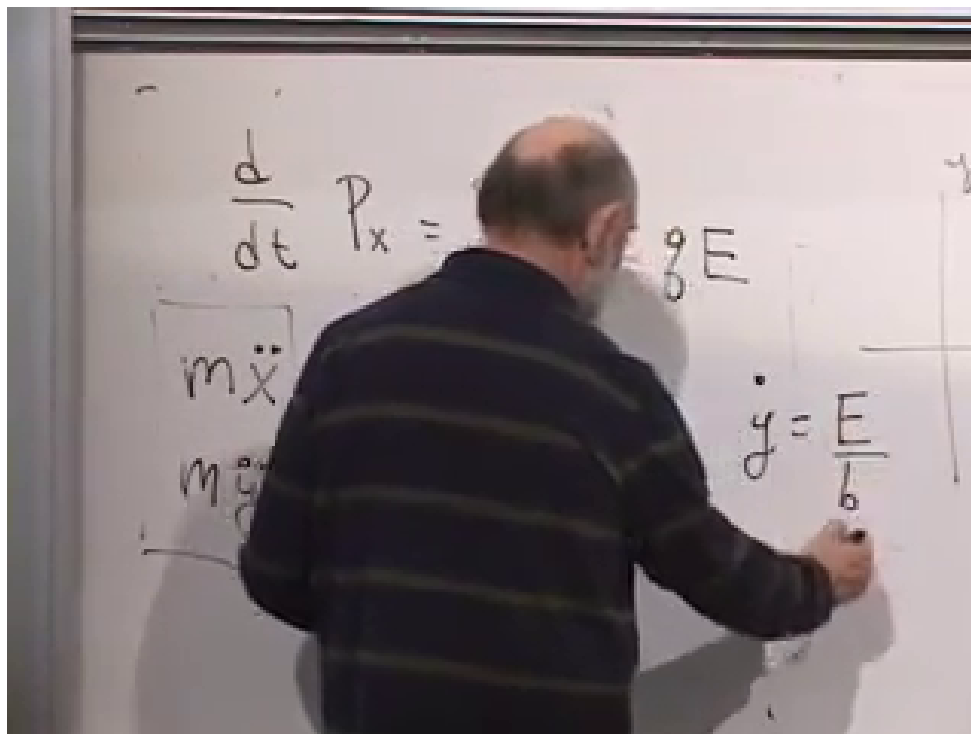


Figure 1.9: The no-acceleration solution at 01:08:20. The electric force points along  $x$ , while the steady drift has  $\dot{y} = E/B$ .

**Question from the audience.** Why does a force applied along  $x$  produce motion along  $y$ ?<sup>a</sup>

**Response.** A trial velocity along  $y$  generates a magnetic force along  $x$ . At the special speed  $E/B$ , that magnetic force exactly cancels the electric force. The net force and acceleration vanish, so the transverse velocity persists.

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<sup>a</sup>Lecture timestamp: 01:07:07–01:10:45.

**Question from the audience.** Is this related to Hall-effect devices used with motors or disk drives?<sup>a</sup>

**Response.** Hall sensors do exploit transverse electromagnetic response and are commonly used to sense magnetic fields, position, or rotor commutation. The detailed engineering of a motor is beyond this mechanics problem, so we should not infer a particular mechanism without examining the device.

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<sup>a</sup>Lecture timestamp: 01:10:47–01:11:38.

The same kind of perpendicular response appears in gyroscopic precession. A torque that one might naively expect to tip a spinning body instead produces motion around a transverse direction. The physical systems differ, but their coupled first-order equations have a closely related structure.

### 1.6.1 Completing the Hamiltonian exercise

The charged-particle Hamiltonian follows directly from the exercise posed in the lecture. From

$$\mathbf{p} = m\mathbf{v} + q\mathbf{A} \tag{1.42}$$

we have

$$\mathbf{v} = \frac{\mathbf{p} - q\mathbf{A}}{m}. \quad (1.43)$$

The Legendre transform gives

$$\begin{aligned} H &= \mathbf{p} \cdot \mathbf{v} - L \\ &= \frac{(\mathbf{p} - q\mathbf{A})^2}{2m} + qV. \end{aligned} \quad (1.44)$$

Thus kinetic energy depends on the gauge-invariant mechanical combination  $\mathbf{p} - q\mathbf{A} = m\mathbf{v}$ , even though Hamilton's equations use the canonical coordinate  $\mathbf{p}$ . Applying those equations reproduces the Lorentz force, as it must.

## 1.7 Poisson Brackets

We now leave the charged-particle example and abstract the structure of Hamiltonian mechanics. For canonical coordinates  $(q_i, p_i)$ , Hamilton's equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \quad (1.45)$$

There are twice as many first-order equations as there are configuration coordinates, in place of the same number of second-order Lagrange equations.

Let  $A(q_i, p_i)$  be any function on phase space with no explicit time dependence. Along a trajectory,

$$\begin{aligned} \dot{A} &= \sum_i \left( \frac{\partial A}{\partial q_i} \dot{q}_i + \frac{\partial A}{\partial p_i} \dot{p}_i \right) \\ &= \sum_i \left( \frac{\partial A}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial H}{\partial q_i} \right). \end{aligned} \quad (1.46)$$

This combination motivates a definition. For any two phase-space functions  $A$  and  $B$ , their Poisson bracket is

$$\boxed{\{A, B\} = \sum_i \left( \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right)}. \quad (1.47)$$

Equation (1.46) is now simply

$$\boxed{\dot{A} = \{A, H\}}. \quad (1.48)$$

If  $A$  has explicit time dependence, the complete statement is

$$\frac{dA}{dt} = \frac{\partial A}{\partial t} + \{A, H\}. \quad (1.49)$$

The bracket formula contains both Hamilton equations. With one canonical pair  $(Q, P)$ ,

$$\dot{Q} = \{Q, H\} = \frac{\partial H}{\partial P}, \quad (1.50)$$

$$\dot{P} = \{P, H\} = -\frac{\partial H}{\partial Q}. \quad (1.51)$$

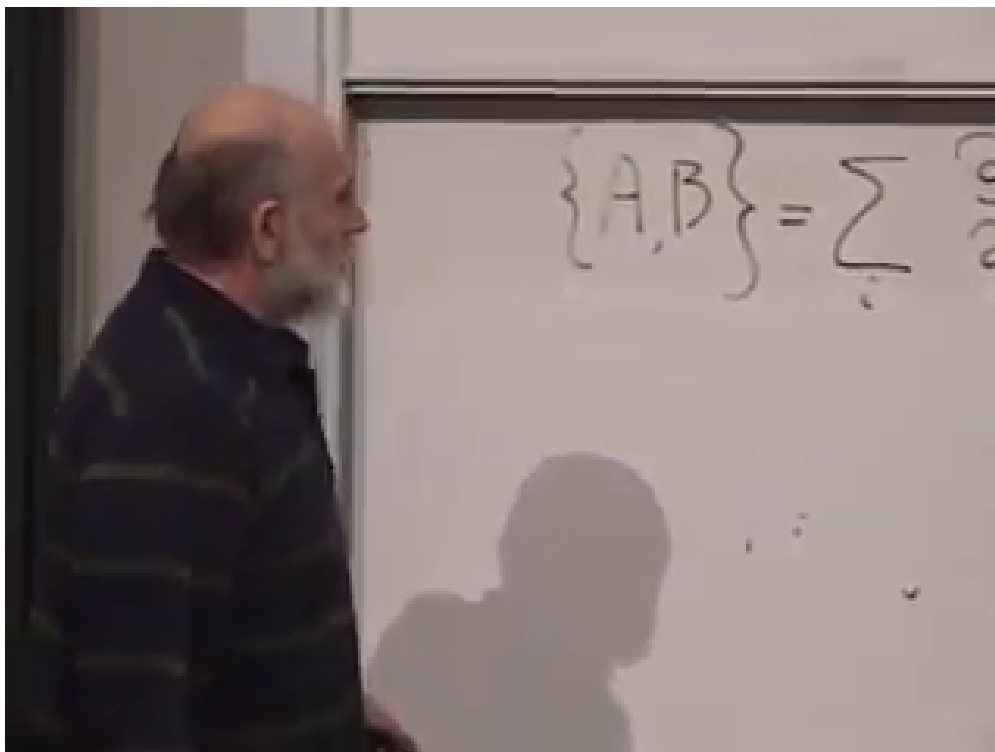


Figure 1.10: The Poisson-bracket definition at 01:20:40. The relative minus sign is the source of its antisymmetry.

So Poisson brackets do not introduce new dynamics. They compress the old dynamics into an algebraic operation on functions.

**Question from the audience.** What happens if  $A$  is the constant function 1, or if it is simply  $P$  or  $Q$ ?<sup>a</sup>

**Response.** A constant has zero derivatives, so  $\{1, H\} = 0$ . Choosing  $A = P$  gives  $\dot{P} = -\partial H / \partial Q$ ; choosing  $A = Q$  gives  $\dot{Q} = \partial H / \partial P$ . The simplest choices recover the two Hamilton equations directly.

<sup>a</sup>Lecture timestamp: 01:22:28–01:25:34.

### 1.7.1 The fundamental algebra

Interchanging the two entries changes the sign:

$$\{A, B\} = -\{B, A\}. \quad (1.52)$$

In particular, any function has zero bracket with itself. For the canonical variables,

$$\boxed{\{q_i, q_j\} = 0, \quad \{p_i, p_j\} = 0, \quad \{q_i, p_j\} = \delta_{ij}.} \quad (1.53)$$

Consequently  $\{p_i, q_j\} = -\delta_{ij}$ : reversing the entries reverses the sign.

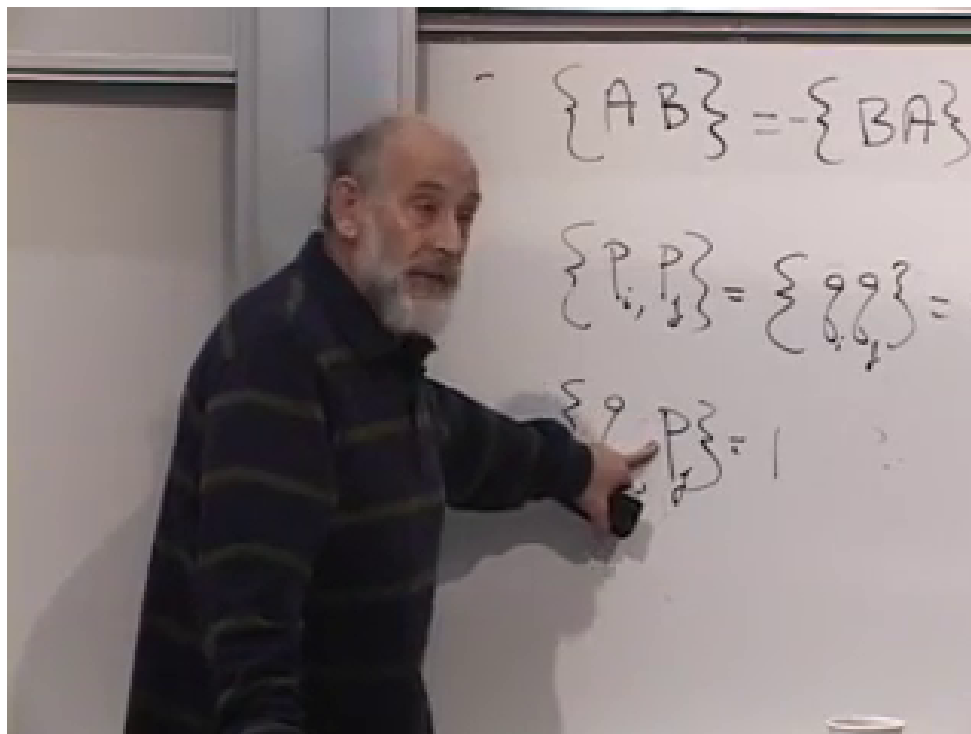


Figure 1.11: Antisymmetry and the fundamental canonical brackets at 01:29:40. The Kronecker delta is one for matching indices and zero otherwise.

**Question from the audience.** Can one see immediately that  $\{P, P\} = 0$ , without expanding the definition?<sup>a</sup>

**Response.** Yes. Antisymmetry says  $\{P, P\} = -\{P, P\}$ . The only quantity equal to its own negative is zero. The same argument gives  $\{Q, Q\} = 0$ .

<sup>a</sup>Lecture timestamp: 01:27:06–01:28:05.

For one canonical pair, bracketing with  $P$  or  $Q$  acts as differentiation:

$$\{P, F\} = -\frac{\partial F}{\partial Q}, \quad \{Q, F\} = \frac{\partial F}{\partial P}. \quad (1.54)$$

More generally,

$$\{p_i, F\} = -\frac{\partial F}{\partial q_i}, \quad \{q_i, F\} = \frac{\partial F}{\partial p_i}. \quad (1.55)$$

This is the symplectic asymmetry of classical mechanics: exchanging a coordinate with its momentum brings in a minus sign.

The bracket is linear in each entry. For constants  $\alpha, \beta$ ,

$$\{\alpha A + \beta C, B\} = \alpha\{A, B\} + \beta\{C, B\}. \quad (1.56)$$



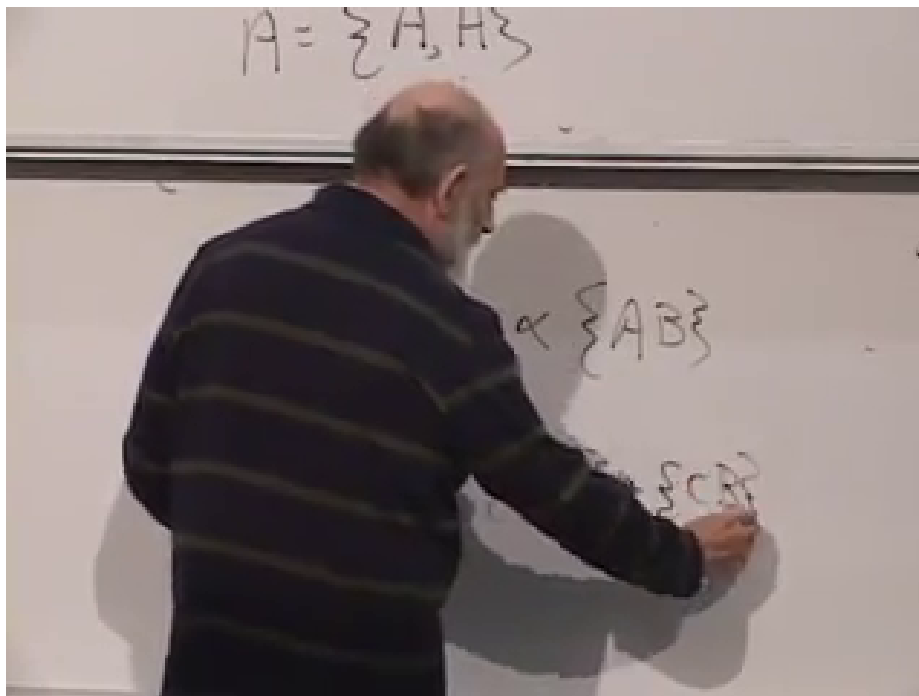


Figure 1.12: Linearity of the bracket at 01:36:20. Multiplication by a constant and addition pass through the bracket.

It also obeys the ordinary product rule in either entry. Expanding the derivatives of  $AB$  gives

$$\begin{aligned}
 \{AB, C\} &= \frac{\partial(AB)}{\partial q_i} \frac{\partial C}{\partial p_i} \\
 &\quad - \frac{\partial(AB)}{\partial p_i} \frac{\partial C}{\partial q_i} \\
 &= A \left( \frac{\partial B}{\partial q_i} \frac{\partial C}{\partial p_i} - \frac{\partial B}{\partial p_i} \frac{\partial C}{\partial q_i} \right) \\
 &\quad + B \left( \frac{\partial A}{\partial q_i} \frac{\partial C}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial C}{\partial q_i} \right) \\
 &= A\{B, C\} + B\{A, C\}.
 \end{aligned} \tag{1.57}$$

Repeated indices are summed in this calculation.

**Question from the audience.** Besides the fundamental brackets, what rules are needed to calculate brackets of general functions?<sup>a</sup>

**Response.** Antisymmetry fixes the order, linearity handles sums and constant multiples, and the product rule handles products. Together with  $\{q_i, p_j\} = \delta_{ij}$ , these rules determine brackets of polynomials and, by suitable limits, the ordinary smooth functions used in mechanics.

<sup>a</sup>Lecture timestamp: 01:33:48–01:40:48.

We can therefore state classical mechanics in a compact way. Begin with canonical variables obeying the fundamental Poisson brackets, specify a Hamiltonian, and evolve every observable by  $\dot{A} = \{A, H\}$ . This formulation follows from Hamilton's equations, but it makes the algebraic structure stand on its own.

The image shows a chalkboard with handwritten mathematical expressions. The top line is partially visible and reads  $\frac{\partial C}{\partial p_i} = \frac{\partial}{\partial p_i} \left( \frac{\partial A}{\partial q_j} B + A \frac{\partial B}{\partial q_j} \right)$ . The main part of the image shows the expansion of the Poisson bracket  $\{AB, C\}$  using the product rule. The expression is written as:

$$\frac{\partial C}{\partial p} - \frac{\partial AB}{\partial q} \frac{\partial C}{\partial q} - A \frac{\partial B}{\partial q} \frac{\partial C}{\partial q} + B \frac{\partial A}{\partial q} \frac{\partial C}{\partial p} - B \frac{\partial A}{\partial q} \frac{\partial C}{\partial q}$$

Figure 1.13: The product-rule expansion at 01:39:00. Grouping the terms yields  $\{AB, C\} = A\{B, C\} + B\{A, C\}$ .

The reason for carrying the abstraction this far is not merely elegance. In quantum mechanics, Poisson brackets are replaced in a precise way by commutators, and the same algebraic pattern becomes the law of motion for operators. Heisenberg and Dirac made that connection central. What looks like an ornate reformulation of classical mechanics here will become one of the shortest routes into quantum dynamics.